Interferometric Scattering (iSCAT) Microscopy

Signal, image and detection limit from first principles,
with 3D tracking, coverslip speckle and photothermal readout

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Abstract

In iSCAT microscopy a sub-wavelength particle on a coverslip scatters a field $E_s \propto a^3$ that interferes with the weak reflection E_r of the glass–water interface, so the camera records $C = 2\,\mathrm{Re}(E_s/E_r) + |E_s/E_r|^2$. We compute this signal from first principles: a Mie series checked against Bohren–Huffman and an independent library; derived leading departures from Rayleigh scattering for both σ_{sca} ($B = -0.200$) and the backscattered amplitude that iSCAT measures ($C = -0.312$); a closed-form scalar image and a vectorial image of a dipole above an interface (Fresnel s/p, aplanatic objective, supercritical light); shot-noise-limited images whose statistics match prediction within error bars; and detection limits for polystyrene, gold and proteins. Six common statements are corrected, among them: the quarter-wave offset is the Gouy phase, not a reflection π shift; the fringe period at the PSF peak is $2\lambda_m/(3+\cos\theta_m)$ (240 nm at NA 1.4, 20% above $\lambda_m/2$); and a particle resting on glass imaged in exact focus has $a_{\min} \propto N^{-1/8}$, not $N^{-1/6}$. Three extensions go further: 3D tracking at the Cramér–Rao bound with MSD analysis, static speckle from coverslip roughness (and why focusing on the interface hides it), and photothermal detection of gold, which beats the speckle floor from $N = 10^9$ photons per pixel.

1 Introduction

Detecting a single unlabelled nanoparticle or protein by its scattered light is hard because scattering scales as a^6. iSCAT [1, 2, 5] turns this into an a^3 signal by interfering the scattered field with a reference, and has enabled label-free single-protein detection and mass photometry [3, 4]. The physics of the contrast (its sign, its dependence on particle height and focus, its noise floor) involves Mie scattering, interface optics and high-NA vectorial imaging together.

This project builds the signal chain from Maxwell's equations to a simulated camera frame and a detection limit, checking each link analytically. It also reuses the Fourier-optics propagator (Gaussian illumination, defocus cross-check), tying the optics projects together.

Parameters. $\lambda_0 = 532\,\mathrm{nm}$; glass $n_1 = 1.52$; water $n_2 = 1.33$ ($\lambda_m = 400\,\mathrm{nm}$); NA 1.4; polystyrene $n = 1.59$; gold $n = 0.544 + 2.231i$ [10].

Contributions.

1. Mie series, and derived x^2 corrections to σ_{sca} and to the backscattered amplitude.
2. Scalar (closed-form) and vectorial (dipole-near-interface) image models, cross-checked.
3. Shot-noise images, PSF fitting and detection limits, protein mass sensitivity, and the effect of attenuating the reference.
4. Extensions: 3D tracking at the Cramér–Rao bound, speckle from coverslip roughness, and photothermal readout.

Table 1: Summary of the main results.

Quantity	Computed	Reference
Mie vs Bohren–Huffman listing ($m = 1.55$, $x = 5.213$)	$Q_{\mathrm{sca}} = 3.10543$, $Q_{\mathrm{back}} = 2.92534$	3.10543, 2.92534
Mie vs independent library	$\leq 4.7 \times 10^{-7}$ (its truncation; ours 5×10^{-12})	—
Rayleigh a^6 law within 2% (polystyrene)	up to 19.99 nm $= \lambda_m/20$; 3.6% at $\lambda_0/20$	—
x^2 correction: σ_{sca} / f_{back}	$B = -0.200$ / $C = -0.312$ ($\to -2/5$ as $m \to 1$)	Eqs. (10), (11)
Log-log slopes 10–40 nm, Mie (polystyrene)	2.927 (interference), 5.855 (scattering)	3, 6 (Rayleigh)
Reference reflectivity	$r = +0.0667$, $\lvert r\rvert^2 = 0.444\%$, no π shift	—
Lossless particle at $h = z_f = 0$	$\arg(E_s/E_r) = \pi/2$ exactly (Gouy)	—
Fringe period, NA 1.4, focus on interface	240.1 nm $= 1.20\,\lambda_m/2$	$\lambda_m/2$ only as NA$\to 0$
Fringe period, spot-summed ($W = 3\,\mu$m)	$1.000451\,\lambda_m/2$	$1.000450\,\lambda_m/2$
Vectorial PSF FWHM along / across $\mathbf{E}_0$	216 / 176 nm (Airy 196)	—
Simulated centre pixel, 2000 images	mean $-2.4127(98) \times 10^{-2}$, std $4.394(69) \times 10^{-3}$	-2.4083×10^{-2}, 4.391×10^{-3}
Smallest radius $N = 10^6$, 3σ, best focus	PS 7.68 nm (5.04 fit); Au 2.63 nm (1.71)	—
$d \ln a_{\min}/d \ln N$	-0.164 best focus; -0.122 focus on interface	$-1/6$; $-1/8$
Protein contrast	-1.83×10^{-3} per MDa (best focus)	—
3D tracking, Au $a = 10$ nm, $N = 10^4$	fit spread = CRLB; $\sigma_z = 0.67$ nm at contrast zero	—
Speckle rms, $\sigma_\varsigma = 0.3$ nm, $\ell = 20$ nm	3.185×10^{-3} Monte Carlo	3.176×10^{-3} theory
Photothermal vs speckle-limited iSCAT (Au)	photothermal wins from $N = 10^9$ at 1 MW/cm^2	—

2 Theory and derivations

2.1 Conventions

Time dependence $e^{-i\omega t}$; lengths in nm. Glass (n_1) fills $z < 0$, water (n_2) fills $z > 0$, and $k_j = 2\pi n_j/\lambda_0$. The particle is a sphere of radius a with centre at height h above the interface ($h = a$ when it rests on the glass), relative index $m = n_p/n_2$ and size parameter $x = k_2 a$. The volume polarizability α is defined by $\mathbf{p} = \varepsilon_0 \varepsilon_2 \alpha \mathbf{E}$ and has units of nm^3. The illumination is an x-polarised plane wave of unit amplitude incident from the glass, and the objective, also in the glass, is focused on the plane $z = z_f$.

2.2 The iSCAT contrast

The camera records the sum of the reference field E_r, reflected at the glass–water interface, and the field E_s scattered by the particle and collected by the objective. Normalising by the particle-free background $|E_r|^2$,

$$C = \frac{|E_r + E_s|^2 - |E_r|^2}{|E_r|^2} = \underbrace{2\,\mathrm{Re}\,\frac{E_s}{E_r}}_{\text{interference}} + \underbrace{\left|\frac{E_s}{E_r}\right|^2}_{\text{scattering}} . \tag{1}$$

For a small particle $|E_s| \ll |E_r|$ and the interference term dominates. It is linear in $E_s \propto \alpha \propto a^3$, which is the origin of the method's sensitivity. The reference is the Fresnel reflection seen from the glass at normal incidence,

$$r = \frac{n_1 - n_2}{n_1 + n_2} = +0.0667, \qquad |r|^2 = 0.444\%. \tag{2}$$

It is positive: there is no π phase shift on reflection from the denser medium when viewed from inside it.

2.3 Rayleigh scattering

Polarizability. A sphere in a uniform field $\mathbf{E}_0$ has a uniform internal field $\mathbf{E}_{\mathrm{in}} = 3\mathbf{E}_0/(m^2 + 2)$ (the solution of Laplace's equation with continuity of $E_\parallel$ and $D_\perp$). Its dipole moment is $\mathbf{p} = \varepsilon_0 \varepsilon_2 (m^2 - 1) V \mathbf{E}_{\mathrm{in}}$, so

$$\alpha = 4\pi a^3 L, \qquad L = \frac{m^2 - 1}{m^2 + 2}. \tag{3}$$

Cross sections. The dipole radiates $E_s = \frac{k^2\alpha}{4\pi} E_0 \sin\psi \, \frac{e^{ikr}}{r}$, with ψ the angle to the dipole axis. Integrating $|E_s|^2 r^2$ over the sphere ($\int \sin^2\psi \, d\Omega = 8\pi/3$) gives the scattering cross section, and the work done by the field on the dipole gives absorption:

$$\sigma_{\mathrm{sca}} = \frac{k^4|\alpha|^2}{6\pi} = \frac{8\pi}{3} k^4 a^6 |L|^2, \qquad \sigma_{\mathrm{abs}} = k \,\mathrm{Im}\,\alpha. \tag{4}$$

Radiation reaction. For real m the static α is real, so $\sigma_{\mathrm{ext}} = \sigma_{\mathrm{abs}} = 0$ while $\sigma_{\mathrm{sca}} > 0$, violating the optical theorem at order α^2. The radiation-reaction form

$$\alpha_{\mathrm{rad}} = \frac{\alpha}{1 - ik^3\alpha/6\pi} \tag{5}$$

restores it exactly: for real α, with $\beta = k^3\alpha/6\pi$, $k\,\mathrm{Im}\,\alpha_{\mathrm{rad}} = k\alpha\beta/(1+\beta^2) = k^4|\alpha_{\mathrm{rad}}|^2/6\pi$.

2.4 Mie scattering and the first corrections

The exact solution for a sphere is the Mie series. The scattering efficiency and the backscattering amplitude are

$$Q_{\mathrm{sca}} = \frac{2}{x^2} \sum_n (2n+1)\big(|a_n|^2 + |b_n|^2\big), \qquad S_1(\pi) = \sum_n \frac{2n+1}{2} (-1)^{n+1} (a_n - b_n). \tag{6}$$

The Mie coefficients. The incident plane wave, the field inside the sphere and the scattered field are each expanded in vector spherical harmonics. The radial functions are spherical Bessel functions inside and outgoing spherical Hankel functions outside. Continuity of the tangential $\mathbf{E}$ and $\mathbf{H}$ at $r = a$ gives, for each multipole order n, two independent 2×2 linear systems, one for the electric (a_n) and one for the magnetic (b_n) multipoles [6]:

$$\begin{aligned} a_n &= \frac{m\,\psi_n(mx)\,\psi_n'(x) - \psi_n(x)\,\psi_n'(mx)}{m\,\psi_n(mx)\,\xi_n'(x) - \xi_n(x)\,\psi_n'(mx)}, \\ b_n &= \frac{\psi_n(mx)\,\psi_n'(x) - m\,\psi_n(x)\,\psi_n'(mx)}{\psi_n(mx)\,\xi_n'(x) - m\,\xi_n(x)\,\psi_n'(mx)}, \end{aligned} \tag{7}$$

with the Riccati–Bessel functions $\psi_n(\rho) = \rho j_n(\rho)$ and $\xi_n(\rho) = \rho h_n^{(1)}(\rho)$. The extinction efficiency is $Q_{\mathrm{ext}} = \frac{2}{x^2}\sum_n (2n+1)\,\mathrm{Re}(a_n + b_n)$, and for real m energy conservation requires $Q_{\mathrm{ext}} = Q_{\mathrm{sca}}$.

For small x the leading coefficients are [6]

$$a_1 = -\frac{2ix^3}{3} L \Big[1 + \frac{3}{5} x^2 \frac{m^2 - 2}{m^2 + 2}\Big] + \frac{4x^6}{9} L^2, \tag{8}$$

$$b_1 = -\frac{ix^5}{45}(m^2 - 1), \qquad a_2 = -\frac{ix^5}{15}\,\frac{m^2 - 1}{2m^2 + 3}. \tag{9}$$

The leading term of a_1 is the Rayleigh dipole; the x^6 term is the radiation reaction.

Scattering cross section. Q_{sca} is a sum of squared moduli with no cross terms. b_1 and a_2 are of order x^5, so their squares enter only at relative order x^4. To order x^2 only the bracket in a_1 matters:

$$\frac{\sigma_{\mathrm{Mie}}}{\sigma_{\mathrm{Ray}}} = 1 + Bx^2, \qquad B = \frac{6}{5}\,\mathrm{Re}\,\frac{m^2 - 2}{m^2 + 2}. \tag{10}$$

Backscattered amplitude. iSCAT measures the field scattered back towards the objective, i.e. $S_1(\pi)$. Here b_1 and a_2 add *coherently* to a_1, so they appear at relative order x^2. Dividing by the dipole amplitude $\frac{3}{2}a_1^{(0)}$,

$$\frac{f_{\mathrm{back}}}{f_{\mathrm{dip}}} = 1 + Cx^2, \qquad C = \frac{3}{5}\,\frac{m^2-2}{m^2+2} - \frac{m^2+2}{30} - \frac{m^2+2}{6(2m^2+3)}. \tag{11}$$

Check: for $m \to 1$, $C \to -\frac{1}{5} - \frac{1}{10} - \frac{1}{10} = -\frac{2}{5}$, which is the Rayleigh–Gans form factor $1 - q^2a^2/10$ at the backscattering momentum transfer $q = 2k$. For polystyrene $C = -0.312$ against $B = -0.200$: the amplitude iSCAT measures leaves the a^3 law sooner than σ_{sca} leaves the a^6 law. The image models use the exact dipole polarizability $\alpha_{\mathrm{Mie}} = 6\pi i a_1/k^3$, which contains both the radiative and dynamic-depolarisation corrections.

2.5 Scalar image

Angular spectrum of a point source. The Weyl expansion writes a spherical wave as a superposition of plane waves,

$$\frac{e^{ikr}}{r} = \frac{i}{2\pi}\iint \frac{e^{i(\boldsymbol{\kappa}\cdot\boldsymbol{\rho} + k_z|z|)}}{k_z}\,d^2\kappa, \qquad k_z = \sqrt{k^2 - \kappa^2}. \tag{12}$$

Image field. The particle is driven by the transmitted field te^{ikh} with $t = 2n_1/(n_1+n_2)$, scatters with amplitude $f = k^2\alpha/4\pi$, and its downward field crosses the interface with $t' = 2n_2/(n_1+n_2)$. The objective forms the image from the angular spectrum at the focal plane, cut at $\kappa_m = k_0 \min(\mathrm{NA}, n_2)$. Performing the azimuthal integral ($\int e^{i\kappa\rho\cos\phi}d\phi = 2\pi J_0(\kappa\rho)$),

$$E_s(\rho) = tt' f\, e^{ikh}\, i\int_0^{\kappa_m} \frac{\kappa\,d\kappa}{k_z}\, J_0(\kappa\rho)\, e^{ik_z d}, \qquad d = h - z_f. \tag{13}$$

Centre of the image. Substitute $u = k_z$, so that $\kappa\,d\kappa/k_z = -du$ and the $1/k_z$ singularity disappears. At $\rho = 0$, $J_0 = 1$ and the integral is elementary:

$$E_s(0) = tt' f\, e^{ikh}\,\frac{e^{ikd} - e^{ikcd}}{d} \xrightarrow{d\to 0} tt' f\, e^{ikh}\, ik(1-c), \qquad c = \cos\theta_m. \tag{14}$$

The quarter-wave offset is the Gouy phase. The reference in the same model is re^{-ikz_f}. At $h = z_f = 0$,

$$\frac{E_s}{E_r} = i\,\frac{tt' f k(1-c)}{r}. \tag{15}$$

For a lossless particle f is real, so E_s/E_r is purely imaginary: the particle in focus is in quadrature with the reference and gives *zero* interference contrast, like a weak phase object in bright-field microscopy. The factor i comes from the focusing integral (the Gouy phase of the scattered wave), not from a reflection phase.

Fringe period with the focus on the interface. For $z_f = 0$, $d = h$ and

$$e^{ikh} - e^{ikch} = 2i\, e^{ik(1+c)h/2}\sin\frac{k(1-c)h}{2}. \tag{16}$$

Combining with the prefactor e^{ikh} and the reference,

$$\arg\frac{E_s(0)}{E_r} = \frac{\pi}{2} + \frac{k(3+c)}{2}\,h \quad\Longrightarrow\quad \Lambda = \frac{2\lambda_m}{3+\cos\theta_m}. \tag{17}$$

Physically, the phase advances by k on the way to the particle and by k_z on the way back, with k_z averaged over the collected angles. $\Lambda = \lambda_m/2$ only as $\mathrm{NA}\to 0$ ($c \to 1$). If instead the focus follows the particle, $z_f = h$, then $E_s/E_r \propto e^{2ikh}$ and the period is exactly $\lambda_m/2$.

Particle resting on the glass. With $h = a$ and $z_f = 0$, the interference term is

$$2\,\mathrm{Re}\,\frac{E_s}{E_r} = -2\left|\frac{tt'f}{r}\right| k(1-c)\sin\frac{k(3+c)a}{2} \approx -\left|\frac{tt'f}{r}\right| k^2(1-c)(3+c)\,a. \tag{18}$$

Because $f \propto a^3$, the contrast scales as a^4 rather than a^3. This changes the detection-limit scaling (Section 2.7).

2.6 Vectorial image

Dipole spectrum. The angular spectrum of a dipole $\mathbf{p} = \varepsilon_0\varepsilon_2\alpha\, t e^{ik_2 h}\hat{x}$ in medium 2 is [8]

$$\mathbf{E} = \frac{ik_2^2}{8\pi^2\varepsilon_0\varepsilon_2}\iint \frac{(\mathbb{1} - \hat{\mathbf{k}}\hat{\mathbf{k}})\,\mathbf{p}}{k_{z2}}\, e^{i(\boldsymbol{\kappa}\cdot\boldsymbol{\rho} + k_{z2}|z-h|)}\, d^2\kappa. \tag{19}$$

For a downward wave with azimuth ϕ the s and p unit vectors are

$$\hat{s} = (-\sin\phi, \cos\phi, 0), \qquad \hat{p} = (\cos\theta_2\cos\phi,\ \cos\theta_2\sin\phi,\ \sin\theta_2), \tag{20}$$

so $\mathbf{p}\cdot\hat{s} = -p\sin\phi$ and $\mathbf{p}\cdot\hat{p} = p\cos\theta_2\cos\phi$.

Fresnel transmission. For an s-polarised plane wave travelling from water into glass, the tangential electric field is continuous, $1 + r_s = t_s$, and so is the tangential magnetic field, which is proportional to k_z times the amplitude: $k_{z2}(1 - r_s) = k_{z1}t_s$. Eliminating r_s gives t_s below. The same two conditions for p polarisation, written for the magnetic field, give t_p.

Transmission into the glass. Each component is transmitted with the Fresnel coefficient

$$t_s = \frac{2k_{z2}}{k_{z2} + k_{z1}}, \qquad t_p = \frac{2n_1 n_2\, k_{z2}}{n_1^2 k_{z2} + n_2^2 k_{z1}}. \tag{21}$$

Both are proportional to k_{z2}, which cancels the $1/k_{z2}$ of the dipole spectrum. The integrand is therefore finite at the critical angle and continues smoothly into the supercritical range $n_2 < \mathrm{NA} < n_1$, where k_{z2} is imaginary and the dipole's near field couples into the glass.

Objective and tube lens. An aplanatic objective obeys the sine condition; energy conservation between a plane-wave component and the pupil gives the apodisation $\sqrt{\cos\theta_1}$. With a paraxial tube lens, $\hat{s} \to \hat{\phi}$ and $\hat{p} \to \hat{\rho}$ in the pupil, and the camera field is a 2D Fourier transform. The azimuthal integrals give Bessel functions of order 0 and 2:

$$\begin{aligned} E_x &= P\int \kappa\, d\kappa\, g(\kappa)\Big[(T_s + \cos\theta_2 T_p)\, J_0(\kappa\rho) + (T_s - \cos\theta_2 T_p)\, J_2(\kappa\rho)\cos 2\phi_\rho\Big], \\ E_y &= P\int \kappa\, d\kappa\, g(\kappa)\,(T_s - \cos\theta_2 T_p)\, J_2(\kappa\rho)\sin 2\phi_\rho, \end{aligned} \tag{22}$$

with

$$P = \frac{ik_2^2\alpha\, t\, e^{ik_2 h}}{8\pi}, \qquad T_{s,p} = \frac{t_{s,p}}{k_{z2}}, \qquad g = \sqrt{\frac{k_{z1}}{k_1}}\, e^{ik_{z2}h - ik_{z1}z_f}. \tag{23}$$

The $J_2\cos 2\phi_\rho$ term elongates the PSF along the polarisation. The reference is the $\kappa = 0$ wave, $re^{-ik_1 z_f}$, and only E_x interferes with it:

$$C = 2\,\mathrm{Re}\,\frac{E_x}{E_r} + \frac{|E_x|^2 + |E_y|^2}{|E_r|^2}. \tag{24}$$

Spot-summed signal. Averaging E_x over the camera with a Gaussian weight $e^{-\rho^2/W^2}$ multiplies the integrand by its transform $\pi W^2 e^{-\kappa^2 W^2/4}$, and the J_2 term averages to zero. Only $\kappa \lesssim 2/W$ survives, where $k_{z2} \approx k_2 - \kappa^2/2k_2$. With $\langle \kappa^2 \rangle = 4/W^2$ under this weight, the phase advances at $2k_2 - 2/(k_2 W^2)$ per unit height, so

$$\Lambda_W = \frac{\lambda_m}{2}\Big(1 + \frac{1}{(k_2 W)^2}\Big), \tag{25}$$

at any NA. The spot-summed signal recovers $\lambda_m/2$; the peak pixel does not.

2.7 Noise and detection

Shot noise of the ratiometric contrast. A pixel at intensity I (in units of $|E_r|^2$) collects Poisson(NI) photons. The contrast is estimated as $\hat{C} = S/\hat{B} - 1$, with the background $\hat{B}$ averaged over n_{bg} particle-free frames. To first order in the fluctuations (delta method),

$$\operatorname{Var}\hat{C} = \frac{I}{NB^2} + \frac{I^2}{N_{\mathrm{bg}} B^3} \xrightarrow{I \approx B = 1} \frac{1}{N}\Big(1 + \frac{n_{\mathrm{f}}}{n_{\mathrm{bg}}}\Big). \tag{26}$$

With one signal and one background frame and the exact intensities this becomes $\sigma_C = \sqrt{(1+C)(2+C)/N}$.

Derivation of the noise formula. Let S be the count in the signal frame, with mean NI, and $\hat{B}$ the background count per frame averaged over n_{bg} frames, with mean NB. Write $S = NI(1+\delta_S)$ and $\hat{B} = NB(1+\delta_B)$. For Poisson counts the relative fluctuations are independent, with variances $1/(NI)$ and $1/(N_{\mathrm{bg}}B)$, where $N_{\mathrm{bg}} = n_{\mathrm{bg}}N$. To first order,

$$\hat{C} + 1 = \frac{I}{B}\big(1 + \delta_S - \delta_B\big), \qquad \operatorname{Var}\hat{C} = \frac{I^2}{B^2}\Big(\frac{1}{NI} + \frac{1}{N_{\mathrm{bg}}B}\Big), \tag{27}$$

which is Eq. (26).

PSF fit. With a known template T (peak 1), the least-squares amplitude is $\hat{A} = \sum T\hat{C} / \sum T^2$, with variance $1/(N\sum T^2)$ for white noise. It beats a single pixel by $\sqrt{\sum T^2}$, which is 3.65 for 60 nm pixels at NA 1.4.

Detection limit. A particle is detected when $|C(a)| = k_\sigma/\sqrt{N}$ (with $k_\sigma = 3$). If $C \propto a^3$,

$$a_{\min} \propto N^{-1/6}. \tag{28}$$

This assumes that the phase of E_s/E_r does not depend on a. For a particle resting on the glass with the focus on the interface, $C \propto a^4$ and $a_{\min} \propto N^{-1/8}$.

Strength of the reference. Let the illumination deliver N_{inc} photons per pixel, so the background is $|r|^2 N_{\mathrm{inc}}$, and write $s = E_s/E_0$ with relative phase ϕ. The single-pixel signal-to-noise ratio is

$$\mathrm{SNR} = \frac{\big||r+s|^2 - |r|^2\big|}{|r+s|}\sqrt{N_{\mathrm{inc}}} \;\to\; \begin{cases} 2|s|\cos\phi\sqrt{N_{\mathrm{inc}}}, & |s| \ll |r|, \\ |s|\sqrt{N_{\mathrm{inc}}}, & |r| \ll |s|. \end{cases} \tag{29}$$

At fixed illumination, attenuating the reference does not change the shot-noise SNR until $|r| \sim |s|$. At a fixed number of *detected* photons (a camera full well), $N_{\mathrm{inc}} = N/|r|^2$ and the SNR grows as $1/|r|$; this is the reason for partial reflectors [4].

2.8 Proteins

For a dilute solution $n = n_m + (dn/dc)\,c$, so $\Delta\varepsilon_r = 2n_m(dn/dc)\,c$. Linearised Clausius–Mossotti, $\Delta\varepsilon_r/\varepsilon_m = \mathcal{N}\alpha$, with number density $\mathcal{N} = cN_A/M$, gives

$$\alpha = \frac{2\,(dn/dc)\,M}{n_m N_A}, \tag{30}$$

which is $462\,\mathrm{nm}^3$ per MDa for $dn/dc = 0.185\,\mathrm{mL/g}$. A homogeneous sphere of density ρ and index $n_m(1+\delta)$, with $\delta = \rho(dn/dc)/n_m$ chosen to give the same dn/dc, has $\alpha_{\mathrm{sphere}}/\alpha_{dn/dc} = 3L/2\delta$. Expanding L in δ,

$$\frac{\alpha_{\mathrm{sphere}}}{\alpha_{dn/dc}} = 1 - \frac{\delta}{6} - \frac{2\delta^2}{9} + O(\delta^3), \tag{31}$$

which is 0.961 for $\rho = 1.35\,\mathrm{g/cm}^3$. In either description the contrast is linear in mass.

2.9 Extension: 3D tracking

Cramér–Rao bound. A frame has independent Poisson pixels with means $NI_i(\boldsymbol{\theta})$, with parameters $\boldsymbol{\theta} = (x_0, y_0, h, s)$, where s scales the polarizability. The log-likelihood is $\sum_i [n_i \ln(NI_i) - NI_i]$, and its expected Hessian gives the Fisher information

$$F_{jk} = N \sum_i \frac{\partial_j I_i\, \partial_k I_i}{I_i}, \qquad \sigma_{\theta_j} \geq \sqrt{(F^{-1})_{jj}}. \tag{32}$$

If s is known, the bound is $[(F_{3\times3})^{-1}]_{jj}$, which by the Schur complement is never larger.

Where the information is. For a weak scatterer $\partial I \approx 2\,\mathrm{Re}(\partial E_s/E_r)$. At a zero crossing of the centre contrast the phase of E_s/E_r sweeps through quadrature, so $\partial_h I$ is largest and σ_z smallest, while the lateral gradient of a spot with almost no contrast is small and σ_x largest. At a fringe extremum the roles swap; there h and s both change the depth of the spot, so knowing s helps σ_z most.

Mean squared displacement. Brownian steps of variance $2D\Delta t$ per axis plus independent localisation errors of variance σ^2 give, per axis,

$$\mathrm{MSD}(n\Delta t) = 2Dn\Delta t + 2\sigma^2. \tag{33}$$

A straight line through the first two lags gives the unbiased estimators

$$\hat{D} = \frac{M_2 - M_1}{2\Delta t}, \qquad \hat{\sigma}^2 = \frac{2M_1 - M_2}{2}, \tag{34}$$

which are close to optimal when $\sigma^2 \sim D\Delta t$ [12].

2.10 Extension: coverslip roughness

Sheet model. Let the interface be at $z = \zeta(x, y)$ with $|\zeta| \ll \lambda$. To first order in ζ, the thin layer between 0 and ζ is glass replacing water: a polarisation sheet with polarizability per unit area

$$\alpha_A(\boldsymbol{\rho}) = (m^2 - 1)\,\zeta(\boldsymbol{\rho}), \qquad m = n_1/n_2. \tag{35}$$

At normal incidence the field in the layer is tangential and continuous, so no depolarisation factor appears.

Transfer function. Writing Eq. (22) as a 2D Fourier integral, $E_x(\boldsymbol{\rho}) = \int \frac{d^2\kappa}{(2\pi)^2} F_x(\boldsymbol{\kappa}) e^{i\boldsymbol{\kappa}\cdot\boldsymbol{\rho}}$ with $F_x = 2\pi P\,[a(\kappa) - b(\kappa)\cos 2\phi_\kappa]$, where a, b are the pupil amplitudes of the J_0 and J_2 terms. A sheet images as the convolution of α_A with this transfer function.

Check: a uniform step. For $\zeta = \zeta_0$ only $\kappa = 0$ survives, where $b = 0$ and $a = 2t'/k_2$. Then $E = 2ik_1\zeta_0 r$, which is the first-order term of $re^{2ik_1\zeta_0}$, the reflection from an interface displaced to $z = \zeta_0$. It is a pure phase.

Speckle statistics. For a stationary Gaussian surface with power spectrum $S(\kappa)$, let $G = (m^2 - 1)F_x/E_r$. Since ζ is real and F_x is even in $\boldsymbol{\kappa}$,

$$\sigma_C^2 = \mathrm{Var}\Big[2\,\mathrm{Re}\,\frac{E}{E_r}\Big] = 4\int \frac{d^2\kappa}{(2\pi)^2}\, S(\kappa)\,[\mathrm{Re}\,G]^2, \qquad S = \pi\ell^2\sigma_\zeta^2 e^{-\kappa^2\ell^2/4}, \tag{36}$$

for a Gaussian correlation function $\sigma_\zeta^2 e^{-\rho^2/\ell^2}$. Three consequences follow.

- For $z_f = 0$ and $\kappa < k_2$, a and b are real while P/E_r is imaginary, so $\mathrm{Re}\,G = 0$. With the focus on the interface the subcritical roughness is a pure phase object, like a particle in focus; intensity speckle comes only from supercritical light.
- For $\ell \ll \lambda$ the spectrum is flat over the pupil, $S \approx \pi\ell^2\sigma_\zeta^2$, so $\sigma_C \propto \sigma_\zeta\ell$.
- For $\ell \gg \lambda$ only $\kappa \to 0$ contributes, where $\mathrm{Re}\,G \propto \kappa^2$ from defocus, giving $\sigma_C \propto \ell^{-2}$.

Ratiometric removal. With $u = E_s/E_r$ and $v = E_{\mathrm{rough}}/E_r$,

$$\frac{|1 + v + u|^2}{|1 + v|^2} - |1 + u|^2 = -2\,\mathrm{Re}(uv) + O(uv^2, u^2v), \tag{37}$$

so dividing by a particle-free frame of the same static surface cancels the speckle down to the product of two small quantities.

2.11 Extension: photothermal readout

Heat diffusion. A pump modulated as $I_1 e^{-i\Omega t}$ deposits $P_1 = \sigma_{\mathrm{abs}} I_1$. Outside the particle $-i\Omega T = D\nabla^2 T$, whose decaying spherical solution is $e^{-\gamma r}/r$ with $\gamma^2 = -i\Omega/D$. The amplitude follows from the heat flux through the surface, $-\kappa\,\partial_r T \cdot 4\pi a^2 = P_1$:

$$T(r) = \frac{P_1}{4\pi\kappa(1 + \gamma a)}\,\frac{e^{-\gamma(r-a)}}{r}, \qquad \gamma = \frac{1 - i}{r_{\mathrm{th}}}, \quad r_{\mathrm{th}} = \sqrt{2D/\Omega}. \tag{38}$$

Thermal lens in the Born approximation. The index change $\delta n = (dn/dT)T$ is weak, so the lens scatters with polarizability $\delta\alpha(\mathbf{q}) = \int(\delta\varepsilon/\varepsilon_m)e^{i\mathbf{q}\cdot\mathbf{r}}d^3r$, with $\delta\varepsilon/\varepsilon_m = 2\delta n/n_m$. The radial integral is elementary:

$$\delta\alpha(q) \propto \Phi(q) = \frac{4\pi}{q}\int_a^\infty e^{-\gamma(r-a)}\sin qr\,dr = 4\pi\,\frac{\cos qa + (\gamma/q)\sin qa}{q^2 + \gamma^2}. \tag{39}$$

In reflection the lens is probed at $q \approx 2k$, where $\Phi \to 4\pi\cos(qa)/q^2$ stays finite as $\Omega \to 0$; the signal is flat up to $\Omega \sim Dq^2$. In transmission $q \approx 0$ and $\Phi = 4\pi(1 + \gamma a)/\gamma^2 \propto 1/\Omega$. For each collected plane wave the momentum transfer is $q^2 = \kappa^2 + (k_2 + k_{z2})^2 = 2k_2(k_2 + k_{z2})$.

Lock-in signal and noise. $\delta\alpha(t)$ is real, so its field has two sidebands, and the demodulated contrast is

$$\delta C = \frac{(E_r + E_s)^*\,\delta E_+ + (E_r + E_s)\,\delta E_-^*}{|E_r|^2}. \tag{40}$$

Demodulating Poisson counts gives variance $2/N$ in each quadrature. For small gold $\sigma_{\mathrm{abs}} \propto a^3$, $T(a) \propto a^2$ and $\Phi(2k)$ is independent of a, so $\delta\alpha \propto a^3$, like α itself: photothermal detection does not beat $N^{-1/6}$. What it gains is *selectivity*, since the non-absorbing glass does not respond.

3 Numerical methods

3.1 Mie series

The coefficients are computed in the logarithmic-derivative form of Bohren and Huffman, which avoids evaluating Bessel functions of the complex argument mx directly. With $D_n(\rho) = \psi_n'(\rho)/\psi_n(\rho)$,

$$a_n = \frac{\big[D_n(mx)/m + n/x\big]\psi_n(x) - \psi_{n-1}(x)}{\big[D_n(mx)/m + n/x\big]\xi_n(x) - \xi_{n-1}(x)}, \qquad b_n = \frac{\big[mD_n(mx) + n/x\big]\psi_n(x) - \psi_{n-1}(x)}{\big[mD_n(mx) + n/x\big]\xi_n(x) - \xi_{n-1}(x)}. \tag{41}$$

D_n is computed by the downward recurrence $D_{n-1}(\rho) = n/\rho - 1/[D_n(\rho) + n/\rho]$, started at $\max(n_{\max}, |mx|) + 16$ with $D = 0$; downward recurrence is stable for complex m and the starting error decays within a few steps. $\psi_n(x)$ and $\chi_n(x)$ follow from the upward recurrence $f_n = \frac{2n-1}{x} f_{n-1} - f_{n-2}$, which is stable for real x up to $n \approx x$, and $\xi_n = \psi_n - i\chi_n$. The series is truncated at $n_{\max} = x + 4.05x^{1/3} + 2$ [7]; adding 60 more terms changes the result by less than 10^{-11}.

3.2 Image integrals

- *Scalar image:* Eq. (13) in the variable $u = k_z$, by Gauss–Legendre quadrature, which matches the closed-form centre value Eq. (14) to 10^{-12}. Defocus is cross-checked by building the same field with the angular-spectrum propagator of the Fourier-optics project.
- *Vectorial image:* the κ integral of Eq. (22) is split at $\kappa = k_2$, with $\kappa = k_2 \sin\theta_2$ below and $\kappa = k_2 \cosh u$ above, so that each piece is smooth across the critical angle; 300 Gauss–Legendre nodes per piece converge to 10^{-10}.
- *Fast imager for tracking:* the pupil sums are tabulated as functions of ρ on a 1 nm grid and interpolated (second-order accurate). Height enters only through the factor $e^{ik_{z2}h}$ in the integrand, so a new height costs one weighted sum (~ 1 ms per image).

3.3 Simulated images and detection

Signal and background frames are drawn as independent Poisson images with means NI and N, and divided to give the ratiometric contrast. The PSF amplitude is estimated by linear least squares against the template. The detection limit is the radius where $|C(a)|$ equals the 3σ threshold, found by Brent's root-finding in $\log a$. The focal position is chosen to make a 5 nm particle darkest.

3.4 Tracking

The Fisher matrix is computed from central finite differences of the model image with respect to each parameter. Positions are estimated by weighted nonlinear least squares (Levenberg–Marquardt), with Poisson weights N/I_i frozen at the starting guess; for $NI_i \gg 1$ this is as efficient as maximum likelihood [11]. Brownian tracks are generated with Gaussian steps of variance $2D\Delta t$, imaged, localised, and analysed with the two-lag MSD estimators.

3.5 Rough surfaces and speckle

Gaussian random surfaces are generated by filtering white Gaussian noise in Fourier space with $\sqrt{S(\kappa)}$, on periodic grids of at least 41 μm. The speckle field is the FFT convolution of $(m^2 - 1)\zeta$ with the vectorial transfer function. The Monte-Carlo variance is compared with Eq. (36), evaluated by quadrature.

3.6 Photothermal signal

The lens polarizability is evaluated from the closed-form $\Phi(q)$ for each collected plane wave and inserted into the pupil sums of Eq. (22); for supercritical waves q^2 is complex and Φ is used by analytic continuation. The two sidebands are imaged separately and combined into the lock-in signal.

Table 2: Convergence of the discretisations.

Discretisation	Measured	Predicted
Mie truncation	change from 60 more terms $< 10^{-11}$	converged
Vectorial pupil quadrature	200 vs 400 nodes: $< 10^{-10}$	spectral
Bessel-table interpolation	order 2.00	2
Roughness FFT box size L	error $\propto L^{-1.48}$	$L^{-3/2}$ ($\rho^{-3/2}$ PSF tail)
Speckle variance on FFT grid	error $\propto \Delta\kappa^{1.44}$	$\Delta\kappa^{3/2}$ (lattice points in a disc)

4 Verification

1. Scattering. The Mie code is checked against a published numerical example (Bohren–Huffman, $m = 1.55$, $x = 5.213$) and against an independent implementation over four indices and $x = 0.05$–25. Three physical identities are tested: $Q_{\text{ext}} = Q_{\text{sca}}$ for real m (energy conservation); the optical theorem; and the extinction paradox ($Q_{\text{ext}} \to 2$ for large spheres). In the small-x limit, the Mie dipole polarizability must reduce to Clausius–Mossotti, the Rayleigh formula must equal the dipole cross section, and the radiation-reaction form must satisfy the optical theorem. The derived coefficients B and C are checked against the exact series at $x = 0.02$–0.04, where higher orders are negligible, and the radii where Rayleigh fails are checked against the x^2 law.

2. Imaging. The scalar quadrature must reproduce the closed form Eq. (14); its low-NA PSF must be the Airy pattern; the lossless in-focus particle must be in exact quadrature; the fixed-focus period must follow Eq. (17), the focus-tracking period must be exactly $\lambda_m/2$, and the spot-summed period must follow Eq. (25). Defocus must agree with the independent Fourier-optics propagator. For the vectorial model: a closed form exists for $n_1 = n_2$, $h = z_f = 0$, from $\int_c^1 (s^{1/2} + s^{3/2})\, ds$; the model must reduce to the scalar one at low NA with $O(\text{NA}^2)$ error; E_x and E_y must have the symmetry of Eq. (22); the PSF must be elongated along the polarisation; the pupil terms must equal the Fresnel coefficients divided by k_z; and the Fresnel coefficients must conserve energy.

3. Noise and detection. Background frames must have noise $1/\sqrt{N}$; the simulated peak contrast and noise must match Eq. (26) within their statistical errors; the template fit must be unbiased with the predicted error; the detection limit must scale as $N^{-1/6}$ at best focus and break this law at fixed focus; and the three regimes of Eq. (29) must be reproduced.

4. Extensions. The tracking fit must reach the Cramér–Rao bound and the bound must be invariant under whole-pixel translation; the MSD fit must recover D and σ. A uniform step must be a pure phase, a single bump must image as a dipole, the speckle variance must match Eq. (36) and its asymptotes, and its statistics must not depend on the random seed. The temperature must solve the heat equation; the lens form factor must match the Born integral; the sideband formula must match direct time-domain demodulation; and the lock-in noise must be $\sqrt{2/N}$.

Negative controls. Every positive claim has a control in which the effect must vanish:

Claim	Control where the effect must vanish
a particle gives contrast	particle-free frames: zero mean, $1/\sqrt{N}$ noise
the interference term carries the signal	lossless particle exactly in focus: zero interference contrast
absorption heats gold	lossless sphere absorbs nothing
roughness makes speckle	flat or index-matched surface; uniform step is pure phase; NA $\leq n_2$ at $z_f = 0$
speckle limits detection	ratio to particle-free frame removes it to second order
the tracker finds particles	particle-free images give zero fitted amplitude within the bound
MSD intercept measures localisation error	noise-free tracks give zero intercept
thermal lens gives a lock-in signal	no absorption or $dn/dT = 0$ gives exactly zero

5 Results

5.1 Mie versus Rayleigh

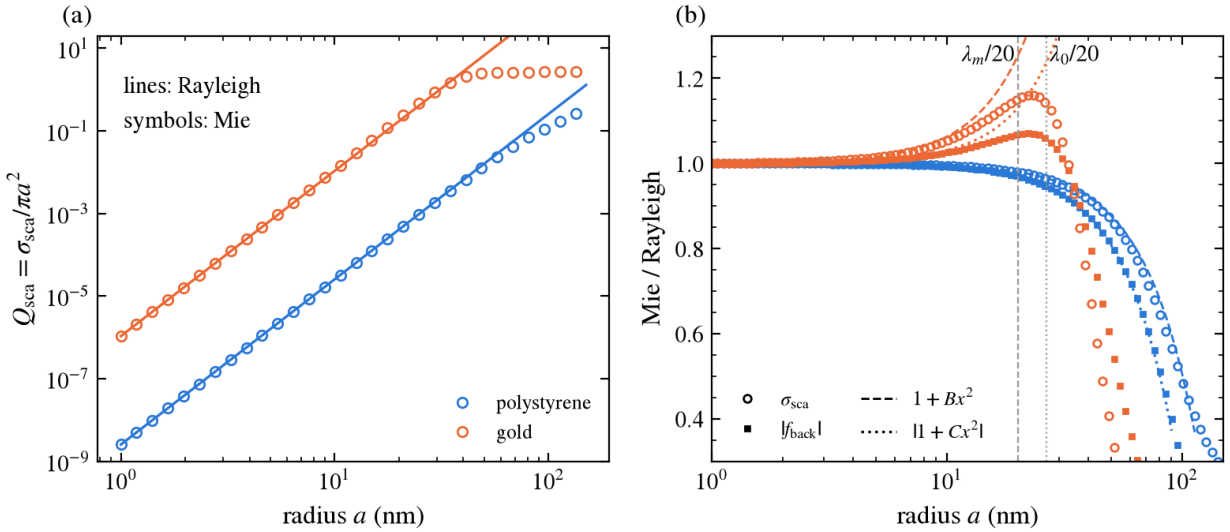


Figure 1: (a) Scattering efficiency, Rayleigh (lines) and Mie (symbols). (b) Mie/Rayleigh ratio for σ_{sca} (circles) and for the backscattered amplitude (squares), with $1 + Bx^2$ (dashed) and $|1 + Cx^2|$ (dotted).

	2% radius (σ)	10% radius (σ)	2% radius ($\lvert f_{back}\rvert$)	10% radius ($\lvert f_{back}\rvert$)
polystyrene	19.99 nm	43.61 nm	16.13 nm	36.09 nm
gold	5.89 nm	14.43 nm	8.78 nm	36.82 nm

The x^2 law predicts the polystyrene 10% radius as 45.0 nm (3% off); for gold, 12.6 nm (14% off) because the radiative x^3 term matters near the plasmon.

5.2 Contrast versus radius

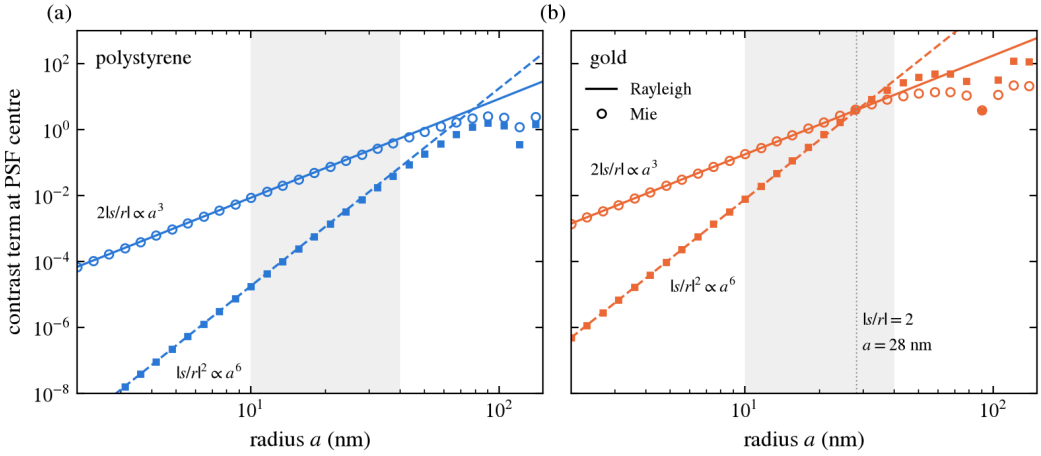


Figure 2: Interference term $2|s/r|$ and scattering term $|s/r|^2$ at the PSF centre ($h = 20\,\text{nm}$, focus on interface, NA 1.4), Rayleigh (lines) and Mie (symbols); 10–40 nm shaded.

For polystyrene (Mie) the slopes are 2.927 and 5.855; the interference term dominates up to 149 nm. Gold ($|L_{\text{Au}}/L_{\text{PS}}| = 20.5$) enters the pure-scattering regime at 28 nm. The scattering slope departs from 6 more than the interference slope departs from 3, because $|C| > |B|$.

5.3 Contrast versus height

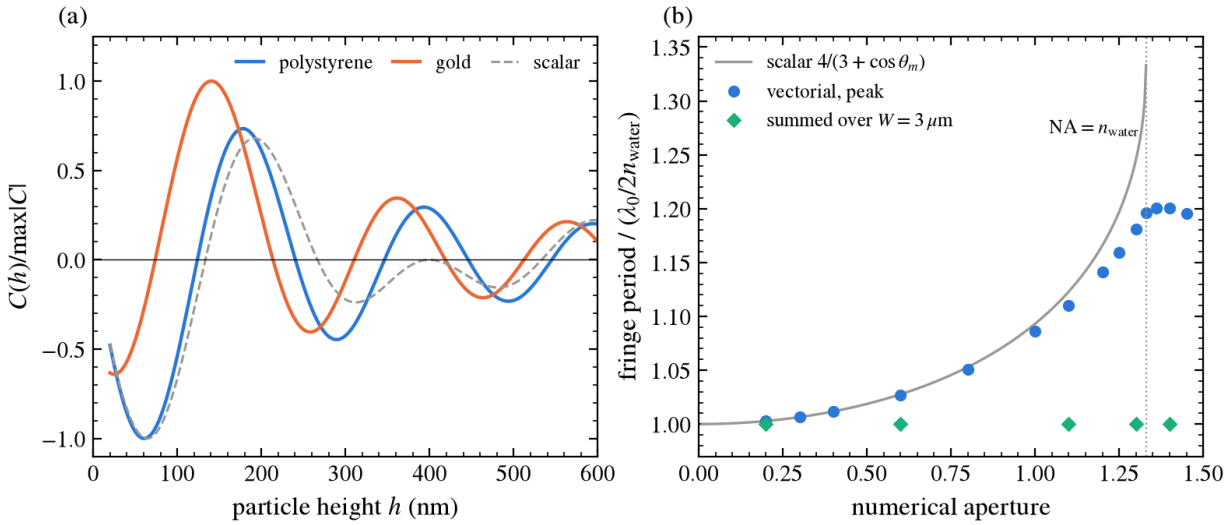


Figure 3: (a) Centre contrast vs height, 20 nm particles, NA 1.4, focus on interface; dashed: scalar model. (b) Fringe period in units of 200 nm: vectorial peak (circles), scalar $4/(3+\cos\theta_m)$ (line), spot-summed (diamonds).

NA	0.2	0.4	0.8	1.0	1.2	1.33	1.4
vectorial peak period (nm)	200.57	202.32	210.18	217.26	228.18	239.17	240.12
scalar $2\lambda_m/(3+c)$ (nm)	200.57	202.34	210.59	218.62	233.15	266.67	—

The period stays within 1% of $\lambda_m/2$ only below $\text{NA} \approx 0.37$. Above $\text{NA} = n_2$ the extra rays are evanescent in water and add no z-phase, so the vectorial period saturates near 240 nm.

5.4 Point-spread function

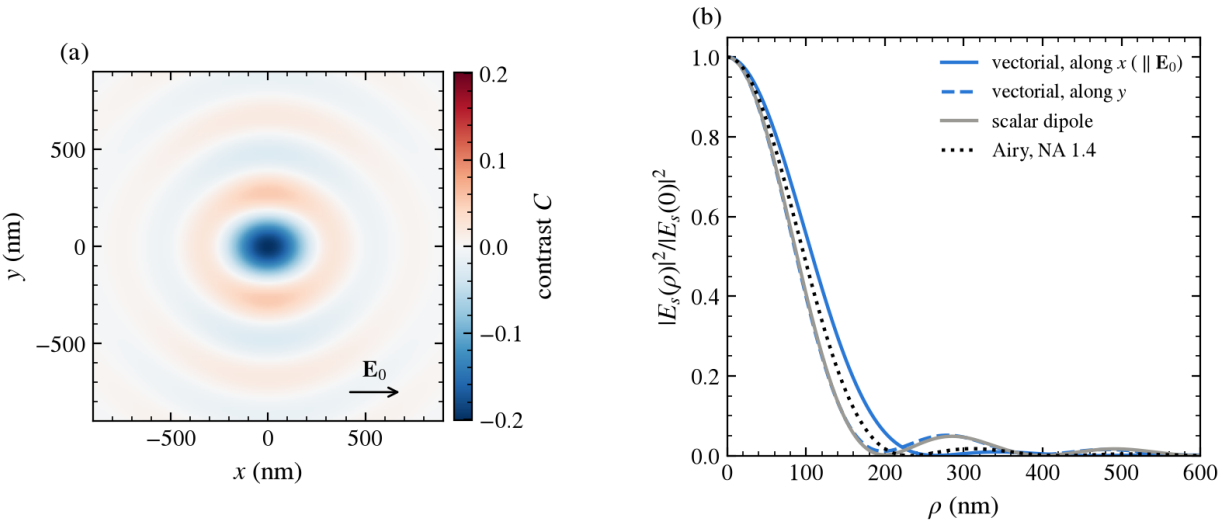


Figure 4: (a) Vectorial iSCAT image of a 30 nm polystyrene sphere at best focus ($z_f = 140$ nm), peak $C = -0.201$. (b) $|E_s|^2$ profiles at $z_f = 0$.

FWHM of $|E_s|^2$: 216.2 nm along $\mathbf{E}_0$, 175.6 nm across (ratio 1.23, from the $J_2 \cos 2\phi$ p-polarised term), scalar dipole 177.1 nm, Airy 195.5 nm. The interferometric central lobe (301 nm) is wider than the intensity PSF because the camera records the field, not its square.

5.5 Simulated images and noise

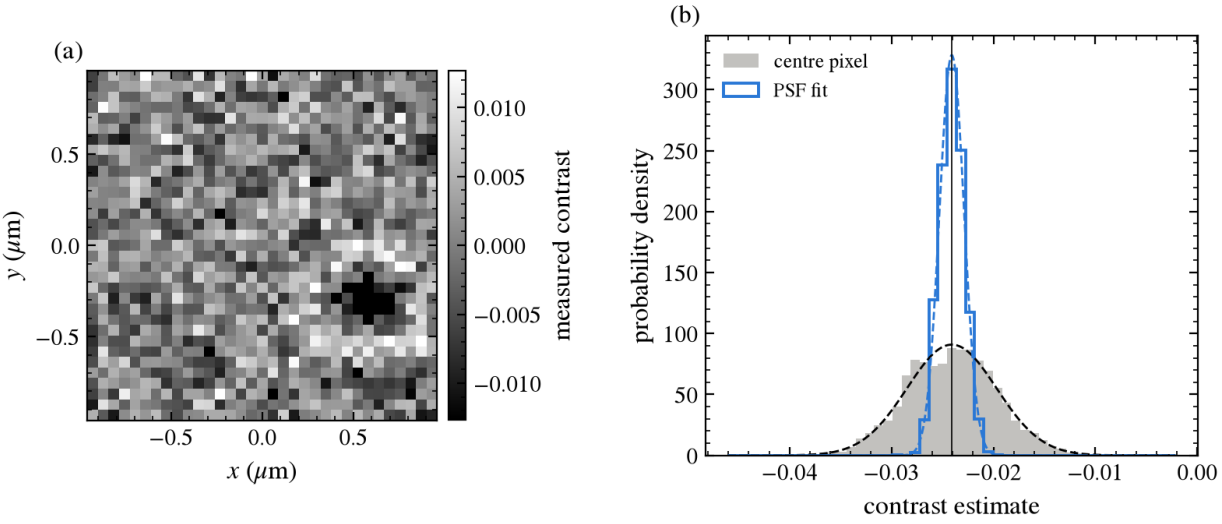


Figure 5: (a) One simulated image: 15 nm polystyrene at $(600, -300)$ nm under a Gaussian beam ($w_0 = 4\,\mu$m), $N = 10^5$. (b) 2000 images: centre pixel (grey) and PSF-fit amplitude (blue) with predicted Gaussians.

	mean	standard deviation
model	-2.4083×10^{-2}	pixel 4.3913×10^{-3}; fit 1.2145×10^{-3}
centre pixel	$-2.4127(98) \times 10^{-2}$	$4.3935(69) \times 10^{-3}$
PSF fit	$-2.4096(27) \times 10^{-2}$	$1.2099(19) \times 10^{-3}$

Every number is within 1σ of the prediction. The background frame raises the pixel noise from $1/\sqrt{N}$ to $\sqrt{(1+C)(2+C)/N}$, and the beam profile cancels in the ratio.

5.6 Detection limit

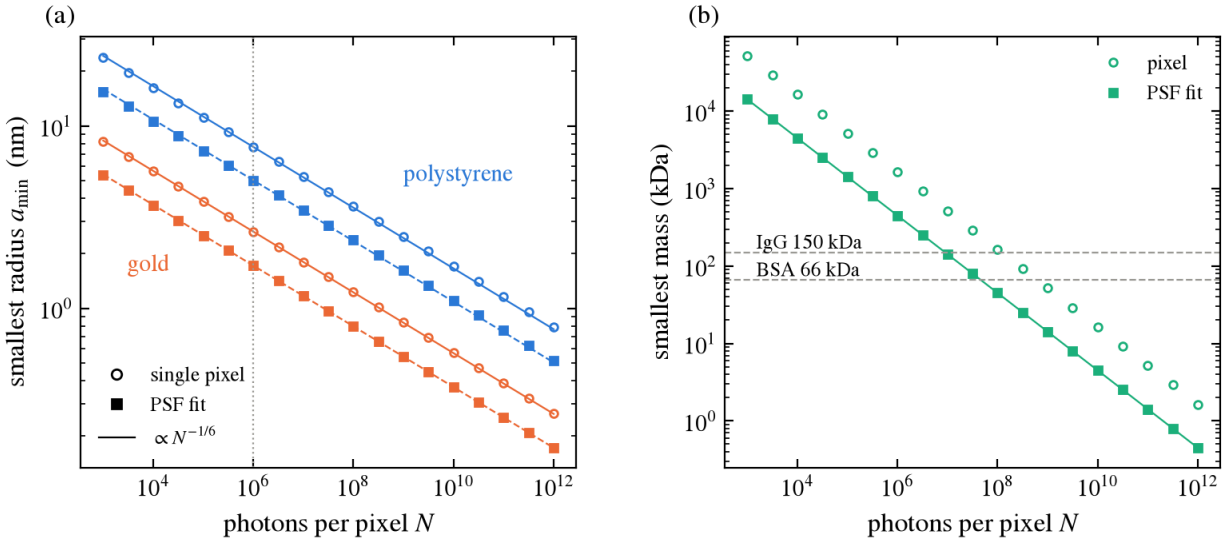


Figure 6: (a) Smallest detectable radius (3σ) vs photons per pixel, single pixel (open) and PSF fit (filled); lines $N^{-1/6}$. (b) Smallest detectable protein mass at $h = 3\,\mathrm{nm}$.

$N = 10^6$	single pixel	PSF fit	slope
polystyrene	7.68 nm	5.04 nm	−0.1644
gold	2.63 nm	1.71 nm	−0.1662
polystyrene, focus on interface	11.23 nm	—	−0.1222

Proteins ($dn/dc = 0.185\,\mathrm{mL/g}$): the smallest detectable mass with a PSF fit is 448, 45 and 4.5 kDa at $N = 10^6, 10^8, 10^{10}$. A 66 kDa protein at 3σ needs 4.6×10^7 photons per pixel, i.e. hundreds of frames at a 10^5 full well, which is why mass photometry averages frames. The equivalent-sphere picture gives $0.962\times$ the dn/dc polarizability, against the derived 0.961.

5.7 Focus and reference strength

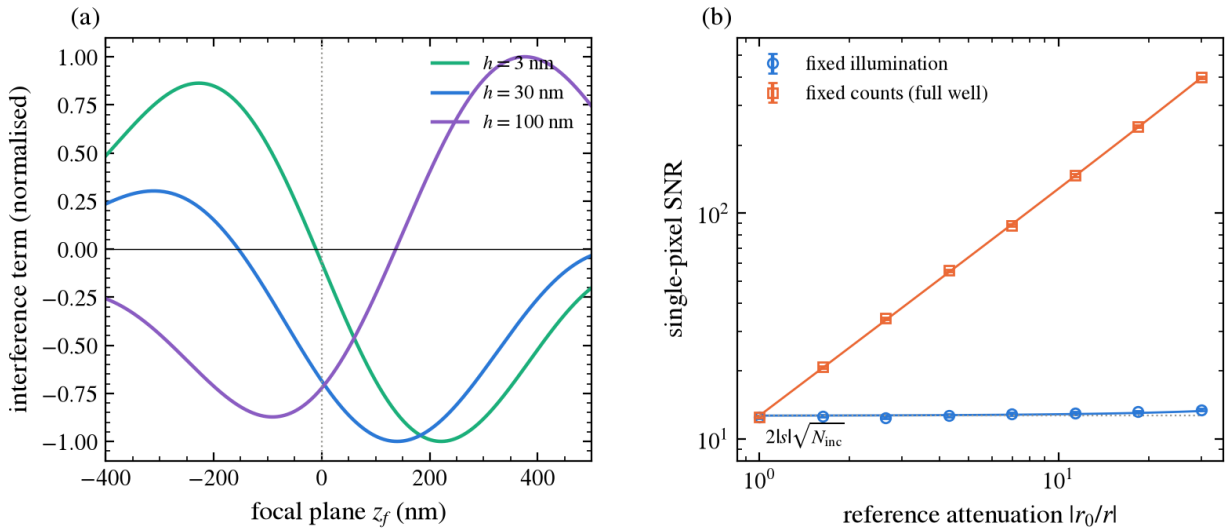


Figure 7: (a) Interference term vs focal position for three heights. (b) Single-pixel SNR vs reference attenuation for fixed illumination (blue) and fixed counts (orange); lines: exact Eq. (29).

Best focus: 221 nm for $h = 3$ nm (14.3× brighter than $z_f = 0$), 140 nm for 30 nm, −91 nm for 100 nm. At fixed illumination the SNR is 12.49(16) at 1× and 13.47(17) at 30× attenuation (theory 12.67, 13.27); at fixed counts it grows as $1/|r|$: 400(5) at 30× (theory 398).

5.8 Extension: 3D tracking

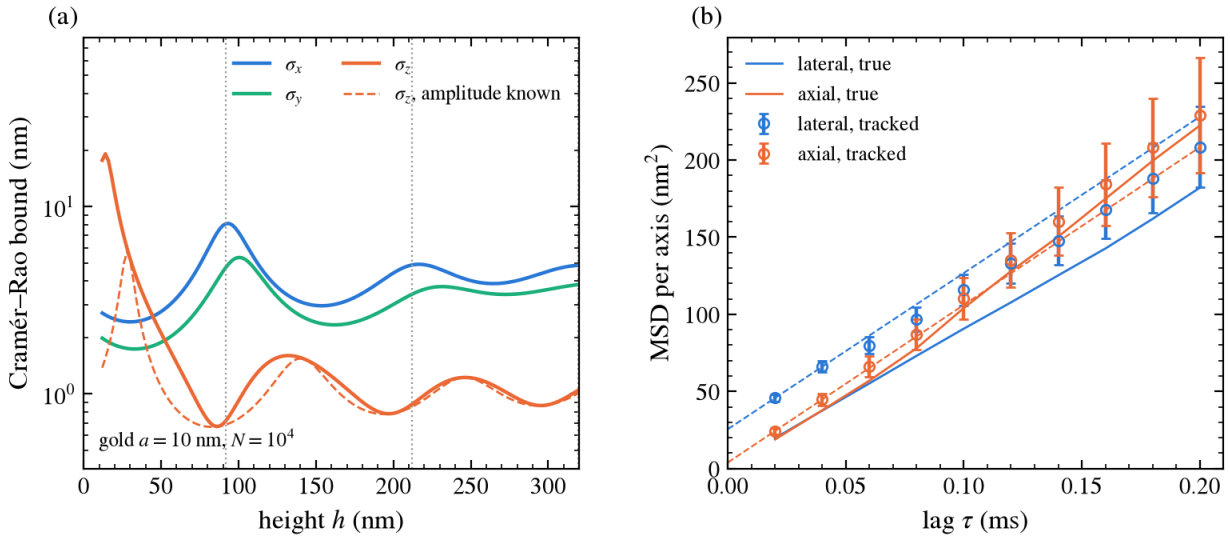


Figure 8: (a) Cramér–Rao bounds for gold $a = 10$ nm, $N = 10^4$, 15 × 15 pixels. (b) Free 3D diffusion ($D = 0.5\,\mu$m^2/s, 50 kHz): MSD of tracked vs true positions, offset $2\sigma^2$.

	$h = 24$ nm (fringe extremum)	$h = 90$ nm (contrast ≈ 0)
σ_x, σ_y	2.45, 1.78 nm	7.94, 4.51 nm
σ_z	8.11 nm	0.69 nm
σ_z with amplitude known	3.91 nm	0.68 nm

At $h = 60\,\mathrm{nm}$ the fit is unbiased and efficient (200 fits: 3.49, 2.28, 1.44 nm vs bound 3.25, 2.14, 1.47 nm). Near the extremum the h–s correlation makes it not yet efficient (+14%). For a 400-frame track, errors are 3.90/2.90/2.51 nm vs bound 4.18/2.84/2.15 nm, and the MSD fit recovers D within its 14% single-track scatter. The fringe ambiguity in h is broken by the PSF shape change at NA 1.4 ($\chi^2 = 746$ vs 232).

5.9 Extension: coverslip speckle

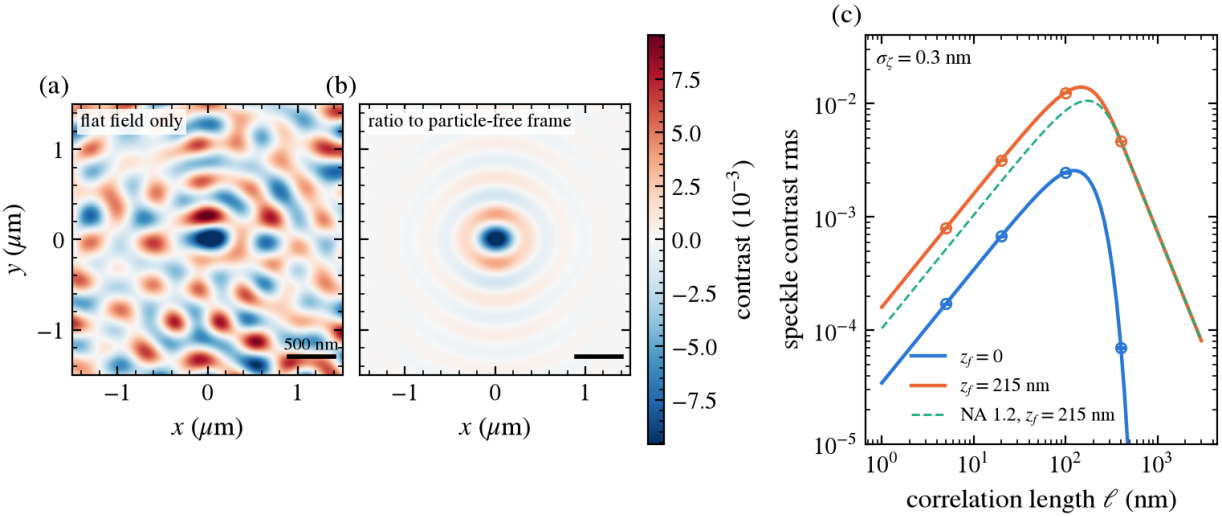


Figure 9: (a) 12 nm polystyrene on glass with $\sigma_\zeta = 0.3\,\mathrm{nm}$, $\ell = 20\,\mathrm{nm}$, at $z_f = 215\,\mathrm{nm}$: speckle rms 3.1×10^{-3}, a quarter of the particle contrast. (b) Divided by a particle-free frame: smooth-glass image to 1.9×10^{-5}. (c) Speckle rms vs ℓ: theory Eq. (36) vs Monte Carlo, within 1.8 standard errors.

	$z_f = 0$	best focus
speckle rms ($\ell = 20\,\mathrm{nm}$)	6.79×10^{-4}	3.18×10^{-3} (215 nm)
ℓ of largest speckle	117 nm	143 nm
3σ floor, polystyrene / gold	10.20 / 2.52 nm	11.16 / 3.41 nm

Speckle is worst at $\ell \approx 100$–$150\,\mathrm{nm}$. Moving the focus amplifies particle and roughness alike, so it does not escape the N-independent floor.

5.10 Extension: photothermal detection

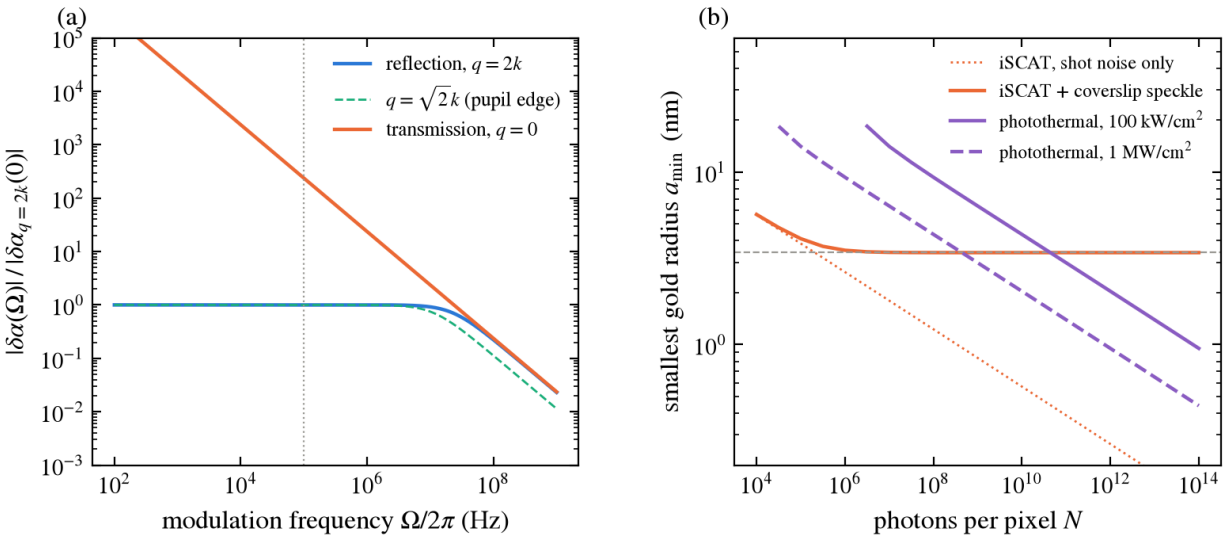


Figure 10: (a) Lens polarizability vs pump modulation frequency for gold $a = 10\,\text{nm}$, normalised to steady reflection. (b) Smallest detectable gold radius: iSCAT shot-noise only, iSCAT with speckle, photothermal at two pump intensities.

gold	σ_{abs}	$\Delta T(a)$ at $100\,\text{kW/cm}^2$	$\delta\alpha(2k)/\alpha$
$a = 5\,\text{nm}$	$44.9\,\text{nm}^2$	$1.19\,\text{K}$	$(-1.97 + 1.95i) \times 10^{-3}$
$a = 10\,\text{nm}$	$374.7\,\text{nm}^2$	$4.97\,\text{K}$	$(-1.88 + 1.98i) \times 10^{-3}$
$a = 20\,\text{nm}$	$3341\,\text{nm}^2$	$22.15\,\text{K}$	$(-1.42 + 2.03i) \times 10^{-3}$

In reflection the signal is within 1% of steady state up to 3.7 MHz (halves at 39 MHz) and is 236× smaller than the forward lens at 100 kHz. At $1\,\text{MW/cm}^2$ photothermal readout goes below the 3.41 nm speckle floor from $N = 10^9$, reaching 0.95 nm at 10^{12}.

6 Discussion

Why volume, not volume squared. The camera measures $|E_r + E_s|^2$; the cross term is linear in $E_s \propto a^3$ and dominates for polystyrene up to $\sim 150\,\text{nm}$.

Optimal reference. Shot noise alone does not set it; the detected-photon budget does. A weaker reference lets more illumination fit in the full well, so the SNR grows as $1/|r|$, limited by photodamage, read noise and the $|s|^2$ term once $|r| \sim |s|$.

Sign changes with height. The phase of E_s/E_r is $\pi/2$ (Gouy) $+ \arg\alpha + (k_2 + \langle k_z \rangle)h$; moving the focus adds its own phase, so the best focus depends on height.

Fundamental limit. Photon shot noise, $1/\sqrt{N}$ per pixel and $1/\sqrt{N \sum T^2}$ for a fit. In practice coverslip speckle often sets the floor first, and only differential (ratiometric) frames remove it.

Approximations and limitations. Homogeneous bulk-index spheres (no surface damping for gold $< 5\,\text{nm}$); normal-incidence plane-wave illumination; image-dipole and near-field local-field corrections neglected (relative size $\sim r(ka)^3$); point dipole with Mie a_1 only (omitted b_1, a_2 change f_{back} by $< 2\%$ up to 16 nm); no aberrations; shot noise only. Photothermal: glass treated

as water for heat flow, gold $d\varepsilon/dT$ ignored, supercritical lens by analytic continuation (a point-dipole comparison brackets the model error at $\sim 1.4\times$). Protein limits were not compared with published mass-photometry data.

7 Conclusion

The iSCAT signal chain (Mie scattering, interface transmission, high-NA vectorial imaging, shot noise) has been built from first principles and checked link by link. The analysis corrects common statements about the reflection phase, the fringe period and the $N^{-1/6}$ law, and derives where the a^3 law first fails for the quantity iSCAT actually measures. The extensions show that 3D tracking reaches the Cramér–Rao bound with the axial and lateral precision trading places along each fringe, that coverslip roughness sets an N-independent floor removable only by differential imaging, and that photothermal readout beats that floor for absorbers.

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